Introduction 1

This course notes introduce ideas and instruments of machine learning form a probabilistic perspective. This approach is growing in importance and is the foundation of the main successes of generative Artificial Intelligence. After a brief discussion on the importance of probability and mathematics, and how to frame machine learning at a high level from this perspective, the structure of the notes will be introduced.

1.1 Models and Probability . . 1
1.1.1 The Importance of Uncertainty 2
1.2 Generative Modelling . . . 3
1.3 Inference and Estimation . 4
1.4 Notes' Structure 4

1.1 Models and Probability

What is Machine Learning all about?

According to Wikipedia, machine learning (ML) explores the study and construction of algorithms that can learn from and make predictions on data. If we dig into this statement, we may wonder what precisely do ML algorithms learn.

The answer to this question is (apparently) simple: they learn **models** of the observed data. Models that can then be used to make predictions or to extract information from such data.

But what is a model?

A model is a hypothesis that certain features of a system of interest are well replicated in another, simpler system. There many possible kinds of models, for instance mental models, verbal models, animal models. We are interested in mathematical ones.

A **mathematical model** is a model where the simpler system consists of a set of mathematical relations between objects (equations, inequalities, etc). We are mostly concerned with **stochastic models**, i.e. a mathematical models where the objects are probability distributions. This is a reasonable and effective choice, as probability is the most effective mathematical theory on the market to deal with uncertainty.

All modelling usually starts by defining a family of models indexed by some parameters, which are tweaked to reflect how well the feature of interest is captured. Machine learning deals with algorithms for automatic selection of one of such models from observations of the system.

In essence, probabilistic machine learning is an umbrella under which we can find several approaches to build probabilistic models of aspects of the world, and several strategies to learn these models from data.

Incorporating probability in the modelling process has the main effect of shifting the focus from point estimates to probability distributions, allowing us to capture uncertainty in the modelling process. It furthermore allows us to account in a more principled way extrinsic noise and intrinsic fluctuations in the data, possibly caused by unknown factors. Additionally, probabilistic modelling is naturally amenable of dealing with limited amount of data (still a crux in certain applications, i.e. in

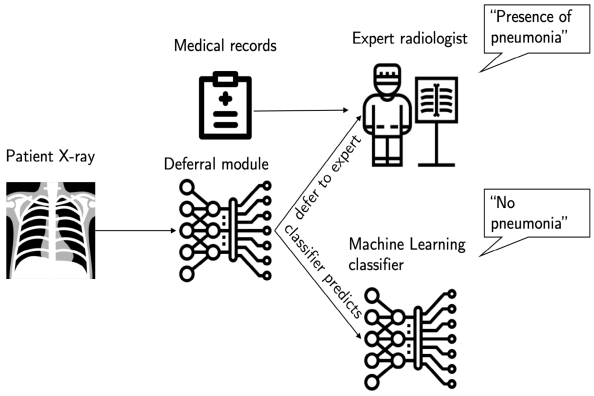

Figure 1.1: A medical scenario in which a network is trained to predict uncertainty of a classifier for medical images and to decide whether to defer decision to an expert radiologist. Taken from https://news.mit.edu/2020/machine-learning-health-care-system-understands-when-to-step-in-0731

medicine). Finally, a probabilistic approach enables us to leverage information and knowledge in a more principled way, either combining multiple models in an ensemble learner, or including prior knowledge in the data using appropriate prior distributions, leading to Bayesian machine learning.

1.1.1 The Importance of Uncertainty

Uncertainty is fundamental when we want to use the results of our models in making decisions. Uncertainty of a prediction is a measure of how much we can trust the prediction itself. Imagine an hypothetical scenario in which we describe uncertainty by Gaussian distribution, centered around the predicted value. Assume the value predicted is 5, but in one case the standard deviation is 0.1 and in the other case is 5. Which prediction would you trust more?

As an example of use of uncertainty to improve decisions, consider the scenario depicted in Figure1.1, in which a deep learning model is analysing x-ray images of patients. The system exploits an uncertainty measure to decide if the answer that will be computed by the model is trustable, and answers only in that case. Otherwise, the decision is deferred to an expert radiologist, that can take also the medical records into account.

Another example related to the safe control of cyber-physical systems is discussed briefly in Figure 1.2.

Uncertainty is broadly coming in two sauces: aleatoric and epistemic. **Aleatoric uncertainty** is essentially related to the randomness of the world around us, which can depend on the level of abstraction that we use in our description. For instance, we can consider a large set of gas molecules in a sealed room. We can in principle describe all their velocities and positions, but such a description will be too complex to obtain and manage. A much simpler and effective description is to assume that the molecules are randomly distributed in the room. This simple intuition gave rise to statistical mechanics in physics, and it is generally a good abstraction to describe complex systems.
Epistemic uncertainty, instead, corresponds to our lack of knowledge about a phenomenon. As an example, consider a pharmaceutical company developing a new drug. They have conducted several clinical trials to

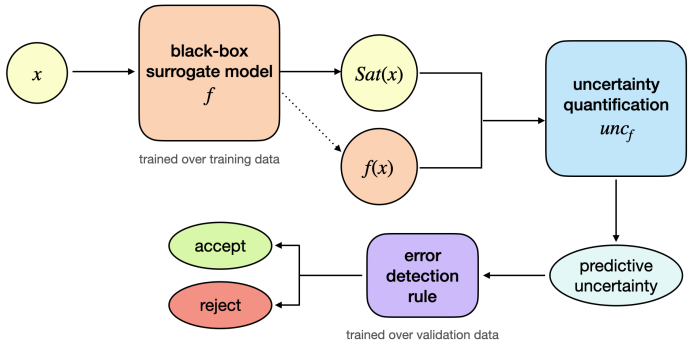

Figure 1.2: Neural predictive monitoring for cyber-physical systems. A neural network predicts is the system will be entering into an unsafe model of operation. The uncertainty of this prediction is then used to detect a potential error in the prediction, and to reject to make predictions that are likely to be wrong. This increases the trustability of the system and allows us to preemptively correct potential failures.
Example taken from L Bortolussi, F Cairoli et al. Neural predictive monitoring, Runtime Verification, 2019.

understand the drug's efficacy and safety profile. However, the drug's long-term effects remain unknown because it has only been tested over short periods. Here, the uncertainty arises from the lack of knowledge about how the drug will affect patients in the long term, not due to random effects. Note that as the drug is used over time in the general population, this epistemic uncertainty can decrease.

1.2 Generative Modelling

We now rephrase the fundamental classification of machine learning tasks in supervised and unsupervised learning using the language of probability. We can distinguish two classes of probabilistic models, which answer to different questions and contain different levels of information. In the following, the variables (features) of a system are divided in two groups: input x and output y. This subdivision is typically related to supervised learning tasks.

Discriminative models aim at describing the conditional probability of the output given the input:

$$p(y \mid x) \text{ or } y = f(x)$$

Instead of the conditional probability, we can restrict to a statistics or a function of $p(y \mid x)$ (for instance, the conditional average of y given x)

Generative models aim at describing the full probability distribution of inputs x or input/output pairs x, y:

$$p(x, y) = p(x)p(y \mid x)$$

Many machine learning tasks can be phrased in this setting:

- **Supervised learning**: learn $p(y \mid x)$ (or the best discriminant $y = f(x)$);
- **Unsupervised learning**: learn $p(x)$ or some properties of it (e.g. clusters);
- **Data generation**: learn a model of $p(x)$ and sample new elements from it (or from $p(x \mid y)$).

Notice that the last task is the one underneath Generative AI applications. For instance GPT-like models generate text in an autoregressive way (i.e. word by word).

1.3 Inference and Estimation

Two central concepts in probability and thus in probabilistic machine learning are the following:

- **Inference**: compute marginals and conditional probability distributions applying the laws of probability.
- **Estimation**: given data and a family of models, find the best parameters/models for the data.

The above distinction is typical in **frequentist** probability, i.e. a formulation of probability theory in which probabilities are seen as fraction of occurrences of events in repeated experiments (at least in principle, not all events are repeatable). The alternative probabilistic formulation is the **Bayesian** one, in which we treat everything as probabilistic, placing a prior distribution on our knowledge (i.e. model parameters) and computing a posterior distribution given the observations. In this framework, estimation boils down to inference.

Note that inference and estimation are typically hard computational tasks, and most of probabilistic machine learning aims at finding clever algorithms to perform efficiently these tasks.

1.4 Notes' Structure

In order to read the following notes and attend with profit classes, basic knowledge of probability, statistics and machine learning is required.

The notes are organized in chapters each dealing with a specific argument. The order of chapters reflects the order of presentation of argument in classes, but also a logical progression from description of models to efficient ways to perform inference and estimation.

Starred chapters and sections contain additional material with respect to the course. You do not have to study these sections, but they can give further insights to the world of probabilistic machine leaning.

Chapter two, however, is a bit anomalous, presenting a formulation of machine learning in terms of **empirical risk minimization** and connecting this formulation to the probabilistic perspective of maximum likelihood. In the chapter, we also discuss some basic idea of **algorithmic learning theory**, namely PAC learning and the related VC dimension and Rademacher complexity.

Chapter three introduces **probabilistic graphical models** (PGM), a graphical language to describe probability distributions (pd). Instances of this family are Bayesian networks and Markov Random Fields. A non-graphical variant of these ideas, using coding construct to model a pd as generative process, give rise to probabilistic programming languages.

One of such languages, Pyro, will be used throughout the course to code probabilistic models.

Chapter four shows how we can exploit the structure of (some) graphical models to perform **inference** tasks in an efficient way. It also discusses limitations of exact inference algorithms like **belief propagation**.

Chapter five discusses a special instance of PGM tailored to temporal series, namely **Hidden Markov Models**.

Chapter six introduces Bayesian machine learning, starting from the simplest model, i.e. **Bayesian linear regression**. We will work out the analytic formulation of regression, and discuss several aspects of Bayesian learning.

Chapter seven digs into the non-analytically tractable Bayesian classification, presenting **Bayesian logistic regression** and one first tool of approximate inference, namely **Laplace approximation**.

Chapter eight presents strategies for approximate inference based on sampling, in particular **Markov Chain Monte Carlo** methods, including some heavily used algorithms like **Hamiltonian Monte Carlo**. We will discuss also how to practically assess convergence of the method.

Chapter nine introduces a method to perform (approximate) parameter estimation for models in which some variables are not observed (i.e. latent). This approach, known as **Expectation Maximization**, is where we will first encounter the **Evidence Lower Bound**, a central tool in modern ML.

Chapter ten discusses the widely used **variational inference**, a method to approximate inference computations which is flexible and can be adapted to stochastic gradient descent algorithms as **Stochastic Variational Inference**. We will also discuss application of this approach for learning **Bayesian Neural Networks**.

Chapter eleven introduces generative modelling, presenting **variational auto-encoders**, one of the mostly used architectures. We will discuss them from the point of view of variational inference, explaining their training procedure in detail.

Chapter twelve continues in presenting a state of the art architecture of generative modelling, especially for images, namely **diffusion models**, in which data is slowly degraded into noise and we learn how to invert such a degradation process.

Chapter thirteen, finally, presents kernel methods and their application to Bayesian learning, namely **Gaussian Processes**, which are a non-parametric approach based on kernels.